

Case Studies CS1 and CS2: Stochastic Lorenz-63 Data Assimilation

4DVarNet-FM

1 State-Space Model

We consider a stochastic Lorenz-63 system augmented with a fourth variable W_L representing a wind forcing. The state vector is

$$\mathbf{x} = [X, Y, Z, W_L]^\top \in \mathbb{R}^4.$$

The deterministic drift is given by

$$\dot{X} = \sigma(Y - X) + c_1 W_L, \quad (1)$$

$$\dot{Y} = X(\rho - Z) - Y, \quad (2)$$

$$\dot{Z} = XY - \beta Z, \quad (3)$$

$$\dot{W}_L = -\gamma(W_L - \bar{W}_L) + c_2 X, \quad (4)$$

with parameters $\sigma = 10.0$, $\rho = 28.0$, $\beta = 8/3$, $\gamma = 0.05$, $\bar{W}_L = 0.0$, $c_1 = 1.0$, $c_2 = 0.1$.

The system is discretised with the Euler–Maruyama scheme using a time step $\Delta t = 0.01$. Multiplicative noise is injected into Y , Z and W_L :

$$X_{t+1} = X_t + \dot{X}_t \Delta t, \quad (5)$$

$$Y_{t+1} = Y_t + \dot{Y}_t \Delta t + \sigma_0 Y_t \sqrt{\Delta t} \xi_1, \quad (6)$$

$$Z_{t+1} = Z_t + \dot{Z}_t \Delta t + \sigma_0 Z_t \sqrt{\Delta t} \xi_2, \quad (7)$$

$$W_{L,t+1} = W_{L,t} + \dot{W}_{L,t} \Delta t + \sigma_L \sqrt{\Delta t} \xi_3, \quad (8)$$

where $\xi_1, \xi_2, \xi_3 \sim \mathcal{N}(0, 1)$ are independent Gaussian increments, $\sigma_0 = 0.08$ and $\sigma_L = 0.20$.

2 Observation Model

Sparse noisy observations of the state variables (X, Y, Z) are available every $\Delta t_{\text{obs}} = 20\Delta t = 0.2$ time units:

$$\mathbf{y}_k = H \mathbf{x}(t_k) + \boldsymbol{\varepsilon}_k, \quad \boldsymbol{\varepsilon}_k \sim \mathcal{N}(\mathbf{0}, R),$$

with $R = R_{\text{var}} I_3$ and $R_{\text{var}} = 0.5$. The observation operator $H \in \mathbb{R}^{3 \times 4}$ selects the first three components. The observation mask is

$$m(t) = \begin{cases} 1 & \text{if } t \in \{\Delta t_{\text{obs}}, 2\Delta t_{\text{obs}}, \dots, T_{\text{max}}\}, \\ 0 & \text{otherwise.} \end{cases}$$

3 Case Study CS1: Perfect Model Baseline

CS1 uses the true dynamics with correct parameters and uncorrupted forcing:

- Parameters: $\sigma = 10.0$, $\rho = 28.0$, $\beta = 8/3$ (no bias).
- Forcing: $W_L^* = W_L$ (the true wind, uncorrupted).
- Coupling in $\dot{X}$: linear term $c_1 W_L$.
- Expected: good reconstruction with $\text{RMSE} \ll 0.5$.

4 Case Study CS2: Model Mismatch

CS2 introduces three sources of model mismatch.

4.1 Parameter Bias

The Lorenz parameters are biased by a fraction $b = 0.15$:

$$\sigma_{\text{CS2}} = \sigma (1 - b) = 8.5, \tag{9}$$

$$\rho_{\text{CS2}} = \rho (1 - b) = 23.8, \tag{10}$$

$$\beta_{\text{CS2}} = \beta (1 + b) \approx 3.11. \tag{11}$$

4.2 Forcing Corruption

The true wind W_L is corrupted by an Ornstein–Uhlenbeck process η and a state-dependent term:

$$W_L^* = W_L + \eta + \alpha X,$$

where $\alpha = 0.15$ (forcing state bias). The corruption η satisfies

$$d\eta = -\frac{1}{\tau_\eta} \eta dt + \sigma_\eta \sqrt{\frac{2}{\tau_\eta}} dW,$$

with $\tau_\eta = 5.0$ and $\sigma_\eta = \sqrt{0.5}$.

4.3 Nonlinear Coupling

The coupling in $\dot{X}$ is changed from linear to quartic:

$$c_1 W_L \longrightarrow \text{sgn}(W_L) W_L^2,$$

which introduces a stronger and non-symmetric forcing on the X variable.

5 Parameter Summary

Table 1: Default model parameters.

Symbol	Value	Unit	Description
σ	10.0	—	Lorenz $X \leftrightarrow Y$ feedback
ρ	28.0	—	Rayleigh number
β	8/3	—	Aspect ratio
γ	0.05	s ⁻¹	Wind relaxation rate
$\overline{W}_L$	0.0	—	Wind baseline
c_1	1.0	—	Wind $\rightarrow$ state coupling
c_2	0.1	—	State $\rightarrow$ wind coupling
σ_0	0.08	—	Multiplicative noise std
σ_L	0.20	—	Wind noise std
τ_η	5.0	s	OU correlation time
σ_η	$\sqrt{0.5}$	—	OU noise std
α	0.15	—	Forcing-state bias
b	0.15	—	Parameter bias fraction
R_{var}	0.5	—	Observation noise variance
B_{var}	2.0	—	Background variance
Δt	0.01	s	Integration step
T_{max}	3.0	s	Assimilation window
Δt_{obs}	0.2	s	Observation interval